

The specific isometry $W = JV$ need not be pure

Candidate full counterexample to the open problem in arXiv:2205.15925

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Status: candidate full counterexample, likely valid. The answer to the open problem following Theorem C of Martin is negative, even for a multiplicity-free diagonal operator. Take $Ae_k = k^{-1}e_k$ on $\ell^2(\mathbb{Z} \setminus \{0\})$, and apply exactly the paper's symmetric-restriction construction using its natural symmetric counting-measure model. If $A = J|A|$, the resulting pure isometry V satisfies

$$JV = I \quad \text{on the infinite-dimensional subspace} \quad \{h : h_{-n} = -h_n\}.$$

Consequently the construction's $W = JV$ has a nonzero (indeed infinite-dimensional) unitary part and is not pure.

1 The exact open problem

Martin proves that if a bounded injective self-adjoint operator A is Hankel with respect to a pure isometry V and $AV > 0$, then, writing $A = J|A|$, the isometry $W = JV$ makes $|A|$ doubly-positive Hankel. The unitary part of W , if present, must be the identity. The subsequent Remark 2 asks whether the *specific* V built in the sufficiency proof from a spectral restriction of A^{-1} might force W to be pure [1]. The source page is reproduced below.

Let $\text{Dom } T$ denote the domain of a linear operator, T . We will say that a closed linear operator, T , is *positive semi-definite* and write $T \geq 0$ if $\langle Tx, x \rangle_{\mathcal{H}} \geq 0$ for all $x \in \text{Dom } T$. If T is also injective, we will write $T > 0$ and say that T is *positive*. Under the additional assumption that $A > 0$, Theorem A recovers a recent result of Niemiec [9, Theorem 1] as a special case:

Corollary B. *Let $A \in \mathcal{L}(\mathcal{H})$ be positive semi-definite and injective with cyclic multiplicity $N \in \mathbb{N} \cup \{+\infty\}$. Then, for any $m \geq N$, $m \in \mathbb{N} \cup \{+\infty\}$, there exists a pure isometry, V , of multiplicity m so that A is V -Hankel and doubly-positive if and only if A is not invertible.*

Remark 1. If $A = A^* \in \mathcal{L}(\mathcal{H})$ is injective, the polar decomposition of A is $J|A|$, where $|A| := \sqrt{A^*A} > 0$ and $J = J^*$ is a symmetry, i.e. a self-adjoint unitary operator. Indeed, if $f(t) = t$ on $[-\|A\|, \|A\|]$ then $f(t) = -\chi_{[-\|A\|, 0]}(t)|t| + \chi_{[0, \|A\|]}(t)|t|$, where χ_Ω denotes the characteristic function of a Borel set, $\Omega \subseteq \mathbb{R}$. By the functional calculus,

$$A = \underbrace{(-\chi_{[-\|A\|, 0]}(A) + \chi_{[0, \|A\|]}(A))}_{=: J} \cdot |A|,$$

and $J = J^* = J^{-1}$ is the difference of two spectral projections. If J is a symmetry then $J^2 = I$ and the spectrum of J is contained in $\{\pm 1\}$. Thus, $\mathcal{H}$ decomposes as $\mathcal{H} = \mathcal{H}_+ \oplus \mathcal{H}_-$ where $Jh_\pm = \pm h_\pm$ for $h_\pm \in \mathcal{H}_\pm$.

Our next result relates the condition that $A = A^*$ is injective, V -Hankel and $AV > 0$ to double-positivity of $|A| := \sqrt{A^*A}$ with respect to $W := JV$ where $A = J|A|$ is the polar decomposition of A .

Theorem C. *Suppose that $A = A^* \in \mathcal{L}(\mathcal{H})$ is injective and Hankel with respect to a pure isometry, V , of multiplicity N and that $AV > 0$. If $A = J|A|$ is the polar decomposition of A , then $|A| > 0$ is W -Hankel and doubly-positive where $W = JV$. If the Wold decomposition of W is $W = W_0 \oplus U$ on $\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}'$ with pure part W_0 and unitary part U , then W_0 has multiplicity N , both $\mathcal{H}_0, \mathcal{H}'$ are $|A|$ -invariant and either $U = I_{\mathcal{H}'}$ or $\mathcal{H}' = \{0\}$.*

Remark 2. It does not seem possible to conclude that the isometry, W , in the statement of the above theorem is pure in general. If, for example, $S = M_z$ is the unilateral shift on H^2 , define a symmetry, J , on H^2 by $J1 = z$, $Jz = 1$ and $Jz^k = z^k$ for all $k \geq 2$. Then, $W = JS$ obeys $W^*1 = S^*J1 = S^*z = 1$, so that W cannot be pure. In the proof of sufficiency in Theorem A given a self-adjoint, injective and non-invertible $A \in \mathcal{L}(\mathcal{H})$, we construct a pure isometry, V , so that A is V -Hankel and $AV > 0$ using a specific restriction of A^{-1} and spectral theory. If $A = J|A|$ is the polar decomposition of A , where J is a symmetry, it may be possible that the specific form of $W = JV$ implies that it is pure in this case. We leave this as an open problem for future research.

Given a self-adjoint, non-invertible and bounded A with trivial kernel, $\text{Ker } A = \{0\}$, we further show that the set of all isometries, V , so that A is V -Hankel are in bijective correspondence with the set of all closed and symmetric restrictions of A^{-1} :

Theorem D. *Let $A \in \mathcal{L}(\mathcal{H})$ be bounded, self-adjoint, injective and not invertible. Then A is Hankel with respect to an isometry, V , with pure part, V_0 of multiplicity N , if and only if there is a closed, symmetric restriction, $[1/A]_0$, of A^{-1} with dense domain in $\mathcal{H}$ and equal deficiency indices (N, N) ,*

$$\dim \text{Ran } ([1/A]_0 \pm iI)^\perp = N,$$

so that $\text{Ran } [1/A]_0 = \text{Ran } V$. Namely, if V is an isometry with pure part of multiplicity N , so that A is V -Hankel, then $[1/A]_0 := A^{-1}|_{\text{Ran } AV}$ is a closed, symmetric restriction of A^{-1} with deficiency indices (N, N) , and if $[1/A]_0$ is a closed, symmetric restriction of A^{-1} with dense domain in $\mathcal{H}$ and deficiency indices (N, N) , then the isometry, V , appearing in the polar decomposition of $[1/A]_0$ has pure part of multiplicity N and A is V -Hankel. If $[1/A]_0$ is simple and $N > 0$ then V is pure, and conversely, if V is pure and if $V^*A = AV > 0$, then $[1/A]_0$ is simple.

The construction first represents A^{-1} as multiplication by the independent variable on $L^2(\mu)$, where μ satisfies the Herglotz condition. It then restricts the multiplication operator to

$$\mathcal{D}_0 = \left\{ h \in \text{Dom } M_t : \int h d\mu = 0 \right\}, \quad T_0 = M_t|_{\mathcal{D}_0}, \quad (1)$$

and takes the isometric polar factor V of T_0 . This defining source page is reproduced next.

Observe that

$$\mu(\mathbb{R}) = +\infty,$$

and that A^{-1} is unitarily equivalent to the self-adjoint multiplication operator M_t , $(M_t f)(t) = t f(t)$ acting on $L^2(\mu) := L^2(\mathbb{R}, \mu)$. Further note that, by construction, μ obeys the *Herglotz condition*,

$$\int_{-\infty}^{\infty} \frac{1}{1+t^2} \mu(dt) < +\infty.$$

Identify A^{-1} with M_t acting on its maximal, dense domain in $L^2(\mu)$, consider the linear manifold,

$$\mathcal{D}_0 := \left\{ h \in \text{Dom } M_t \mid \int_{-\infty}^{\infty} h(t) \mu(dt) = 0 \right\},$$

and set $[1/A]_0 := A^{-1}|_{\mathcal{D}_0}$. By [2, Proposition 3.5, Lemma 3.6] and the discussion following their proofs, $[1/A]_0$ is a closed, injective and simple symmetric operator with deficiency indices $(1, 1)$, and $\text{Dom } [1/A]_0 = \mathcal{D}_0$ is dense in $\mathcal{H}$. We refer the reader to [1, Sections 78–81] and [11, Chapter X.1] for an introduction to the theory of unbounded symmetric operators and their self-adjoint extensions.

2.4 Positive semi-definite quadratic forms

A sesquilinear (or quadratic) form, $q : \text{Dom } q \times \text{Dom } q \rightarrow \mathbb{C}$, where the *form domain* of q , $\text{Dom } q \subseteq \mathcal{H}$, is dense in a Hilbert space, $\mathcal{H}$, is *positive semi-definite* if $q(h, h) \geq 0$ for all $h \in \text{Dom } q$. (Technically, the domain of q is $\text{Dom } q \times \text{Dom } q$ where $\text{Dom } q$ is the form domain of q .) Given such a positive semi-definite quadratic form, q , with dense form domain $\text{Dom } q \subseteq \mathcal{H}$, we say that q is *closed* if its form domain, $\text{Dom } q$, is complete with respect to the norm induced by the inner product,

$$q(x, y) + \langle x, y \rangle_{\mathcal{H}}.$$

That is, q is closed if its form domain is a Hilbert space with respect to the above inner product. A positive semi-definite quadratic form is *closeable* if it has a closed *extension*. Here, as for linear operators, given two positive semi-definite quadratic forms q, q' with dense form domains $\text{Dom } q \subseteq \mathcal{J}$ and $\text{Dom } q' \subseteq \mathcal{H}$, where $\mathcal{J}$ is a closed subspace of $\mathcal{H}$, we say that q is a *restriction* of q' , or that q' is an *extension* of q , and we write $q \subseteq q'$, if $\text{Dom } q \subseteq \text{Dom } q'$ and $q = q'|_{\text{Dom } q \times \text{Dom } q}$. Any closeable, positive semi-definite quadratic form with dense form domain in $\mathcal{H}$ has a minimal closed extension, $\overline{q}$, with dense form domain $\text{Dom } \overline{q} \subseteq \mathcal{H}$. Standard references for the theory of unbounded sesquilinear forms include [10, Chapter VIII.6] and [7, Chapter 6].

A positive semi-definite quadratic form q , with dense form domain in a Hilbert space, $\mathcal{H}$, is closed if and only if there is a unique self-adjoint and positive semi-definite operator, A , with dense domain in $\mathcal{H}$ so that $\text{Dom } q = \text{Dom } \sqrt{A}$ and

$$q(g, h) = q_A(g, h) := \langle \sqrt{A}g, \sqrt{A}h \rangle_{\mathcal{H}}; \quad g, h \in \text{Dom } q,$$

[7, Chapter VI, Theorem 2.1, Theorem 2.23]. This can be viewed as an extension of the Riesz representation lemma for bounded positive semi-definite quadratic forms. The following definition is taken from [7, Chapter VI].

Definition 1. (A partial order on positive semi-definite forms) If q_1, q_2 are densely-defined and positive semi-definite quadratic forms in a Hilbert space, $\mathcal{H}$, we say $q_1 \leq q_2$ if

1. $\text{Dom } q_2 \subseteq \text{Dom } q_1$, and
2. $q_1(x, x) \leq q_2(x, x)$ for all $x \in \text{Dom } q_2$.

If A_1, A_2 are self-adjoint and positive semi-definite operators with dense domains in a Hilbert space, $\mathcal{H}$, we say that $A_1 \leq A_2$ if $q_{A_1} \leq q_{A_2}$. That is, $A_1 \leq A_2$ if

2 A symmetric diagonal model

Let

$$\mathcal{H} = \ell^2(\mathbb{Z}_*), \quad \mathbb{Z}_* := \mathbb{Z} \setminus \{0\},$$

with canonical basis $(e_k)_{k \in \mathbb{Z}_*}$, and define

$$Ae_k = \frac{1}{k}e_k. \quad (2)$$

Then A is bounded, injective, self-adjoint, and not invertible. Its eigenvalues are all simple and tend to zero. It is multiplicity-free: for example,

$$\xi_k = (1 + k^2)^{-1/2} \quad (3)$$

is cyclic. Indeed, each spectral point $1/k$ is isolated in $\{0\} \cup \{1/j : j \in \mathbb{Z}_*\}$; a continuous function on this compact set can select any one coordinate, and polynomial approximation then puts every e_k in the cyclic subspace generated by ξ .

For this cyclic vector, the measure used in the paper's construction has mass

$$(1 + k^2)|\xi_k|^2 = 1$$

at k . Thus its spectral representation is precisely counting measure on $\mathbb{Z}_*$, and

$$T = A^{-1} = M_k, \quad \text{Dom } T = \left\{ h : \sum_{k \neq 0} k^2 |h_k|^2 < \infty \right\}. \quad (4)$$

The Herglotz condition holds because $\sum_{k \neq 0} (1 + k^2)^{-1} < \infty$, while the total measure is infinite. Moreover every $h \in \text{Dom } T$ is absolutely summable, by Cauchy–Schwarz. Consequently (1) becomes exactly

$$\mathcal{D}_0 = \left\{ h \in \text{Dom } T : \sum_{k \neq 0} h_k = 0 \right\}, \quad T_0 = T|_{\mathcal{D}_0}. \quad (5)$$

This is not an alternative restriction: it is the restriction specified in the proof under the cyclic model (3).

3 The parity splitting

Let R be reflection, $(Rh)_k = h_{-k}$, and set

$$\mathcal{H}_e = \ker(R - I), \quad \mathcal{H}_o = \ker(R + I).$$

The odd space $\mathcal{H}_o$ is infinite-dimensional. Its elements have pairwise cancelling coordinates, so

$$\text{Dom } T \cap \mathcal{H}_o \subset \mathcal{D}_0. \quad (6)$$

In fact the form domain splits as

$$\mathcal{D}_0 = (\text{Dom } T \cap \mathcal{H}_o) \oplus (\mathcal{D}_0 \cap \mathcal{H}_e). \quad (7)$$

Multiplication by k reverses parity, whereas multiplication by $|k|$ preserves it. Hence

$$T(\text{Dom } T \cap \mathcal{H}_o) \subset \mathcal{H}_e, \quad T(\text{Dom } T \cap \mathcal{H}_e) \subset \mathcal{H}_o.$$

It follows that the closed quadratic form

$$q_0(h, g) = \langle T_0 h, T_0 g \rangle, \quad h, g \in \mathcal{D}_0, \quad (8)$$

is the orthogonal sum of its odd and even restrictions. On the odd summand, it is simply

$$q_0(h, g) = \langle |T|h, |T|g \rangle.$$

By uniqueness in the representation theorem for closed positive forms, this proves the operator identity

$$|T_0|_{\mathcal{H}_o} = |T|_{\mathcal{H}_o}, \quad (9)$$

with equality of the corresponding domains.

4 A fixed infinite-dimensional subspace of W

The polar decomposition used by the source is

$$T_0 = V|T_0|.$$

The paper proves that T_0 is simple with deficiency indices $(1, 1)$, and its sufficiency argument therefore makes V a pure isometry of multiplicity one with $AV > 0$. We now compute the derived $W = JV$.

Here

$$J e_k = \operatorname{sgn}(k) e_k, \quad |T| e_k = |k| e_k, \quad T = J|T|.$$

The map $|T| : \operatorname{Dom} T \cap \mathcal{H}_o \rightarrow \mathcal{H}_o$ is onto, with bounded inverse $|A|_{\mathcal{H}_o}$. Given any $y \in \mathcal{H}_o$, choose $h = |T|^{-1}y$. Equations (6) and (9), followed by the two polar decompositions, give

$$V y = V|T_0|h = T_0 h = T h = J|T|h = J y.$$

Therefore

$$W y = J V y = J^2 y = y \quad (y \in \mathcal{H}_o). \quad (10)$$

Since W is an isometry, (10) also gives $W^* y = y$; thus $\mathcal{H}_o$ is a reducing subspace on which W is the identity. A pure isometry has $W^{*n} \rightarrow 0$ strongly, which is impossible on any nonzero fixed vector. Hence W is not pure. In fact its identity unitary part is infinite-dimensional, exactly of the type allowed by Theorem C.

This example uses every defining step of the paper's special construction, so it answers the stated open problem without weakening "specific form" to an arbitrary pair (J, V) .

References

- [1] R. T. W. Martin, *On unitary equivalence to a self-adjoint or doubly-positive Hankel operator*, Concrete Operators **9** (2022), 114–126, [doi:10.1515/conop-2022-0132](https://doi.org/10.1515/conop-2022-0132); arXiv:2205.15925.